Course Syllabus

Welcome to CS 3000 Algorithms and Data Summer 2 2022!

This ***asynchronous*** course begins ***Tuesday, July 5th*** and concludes on ***Thursday, August 18th***. Recordings of lectures will be generated from Monday through Thursday (four lecture recordings per week, for a total of seven weeks) and posted alongside any notes and summaries by the instructor. Given that this is an asynchronous course, please watch the recordings so that you learn at a reasonable pace. You will have access to lecture recordings, instructor notes, and topic summaries. You may be given other reading material that can be referenced, especially from the books/slides included below. *We will be using Canvas, Teams, Piazza, and Gradescope during this summer session. Please make sure you have access to all of these platforms to ensure you are not missing important course information*.

Course Outcomes

- Work on a range of computational problems that arise in diverse applications
- Learn new algorithmic design techniques used to solve the problems
- Learn proof techniques critical for reasoning about your algorithms
- Learn analysis techniques critical to determine the efficiency of algorithms
- Learn to enjoy the subtleties and brilliance of algorithmic ideas underlying smart solutions... (those so called "AHA!" moments of understanding or creating smart solutions)

Course Staff

Professor	Pandurangan Chandrasekaran (Pandu)	pa.chandrasekaran@northeastern.edu
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TA	William Cutler	cutler.wi@northeastern.edu

Office Hours

- Every student will be assigned a TA from the list above to make sure you are all on track with the course. You may interact with the corresponding TA in their office hours if you have any question(s) on the topics covered in the class. Your TA will also grade your homework assignment and exam submissions. *This will all be coordinated via the course's Teams page, so please make sure you can access it.*
- Pandu's office hours are **every Thursday, 9:30am EST to 10:30am EST** and will be *via Zoom*.

The link will be provided soon.

Topics and Timeline (https://northeastern.instructure.com/courses/110667/files/15994470?wrap=1)

Books

The following references are useful. The lecture recordings and suggested reading materials from these and other sources would form the core part of the course coverage.

- Cormen, Leiserson, Rivest, Stein, Introduction to Algorithms, 3rd edition. (4th edition is just out but the parts we are referring are nearly identical in both editions). Slides based on this book are available **here (https://northeastern.instructure.com/courses/110667/files/16005180?wrap=1)** . NU Library also offers free online copies of CLRS 3rd edition too.
- Jon Kleinberg, Eva Tardos. Algorithm design, 1st edition. Slides based on this book from the publishers are available **here (https://www.cs.princeton.edu/~wayne/kleinberg-tardos/pearson/)** .
- Jeff Erickson. Algorithms. Following his very generous invitation, please find the book **here (http://jeffe.cs.illinois.edu/teaching/algorithms/)** .

Latex Resources *(*not required for you to use, but provided if you prefer to use it)

- **The Not So Short Introduction to Latex (https://tobi.oetiker.ch/lshort/lshort.pdf)**
- **Latex Tutorials (https://tug.org/twg/mactex/tutorials/ltxprimer-1.0.pdf)** by a user group
- **Overleaf (https://www.overleaf.com/)** (a web-based latex system; not needing to install your own)

Course Policies

There will be 6 homework assignments with a deadline of submission within **three days/72 hours (out Thursdays and due Sundays by 11:59pm EST)** and a **50% penalty for late submissions that must be turned in by the following two days/48 hours to be accepted.** *No homework assignment will be accepted after this time period*. **All extenuating circumstances will be handled by case. Workload from other courses, personal travel, etc. are NOT considered extenuating so please plan accordingly.**

Homework assignments and exams will be found in the "Assignments" section of Canvas and will be submitted on Gradescope. You may handwrite or type your homework assignments and exams, but please generate your work as a pdf file when uploading and make sure to tag your pages on Gradescope.

Exams: There will be two exams during summer 2, a midterm and a final. Both exams will be a

"take-home" exam lasting from **Friday to Sunday**.

- **Midterm exam:** from Friday, July 29th (9am EST) to Sunday, July 31st (11:59pm EST).
- **Final exam:** from Friday, August 19th (9am EST) to Sunday, August 21st (11:59pm EST).

Academic Integrity: *In this course, you will have access to lecture recordings, instructor notes, topic summaries, an assigned TA for course support, an updated course calendar/summary, Piazza support from the instructor, TA's, and other students, as well as university tutoring (**https://undergraduate.northeastern.edu/peer-tutoring/find-a-peer-tutor/** (https://undergraduate.northeastern.edu/peer-tutoring/find-a-peer-tutor/)) to help you succeed. With this, do not read/reproduce solutions that may be available on the internet/that are not your own. Submissions must be individual. Academic integrity policies of the university on copying/plagiarism must be strictly adhered to: no copying and no plagiarism. You may discuss on the solutions with your peers, but write down everything in your own words/terms. Students who copy, plagiarize, or cheat will be reported to the Office of Student Conduct and Conflict Resolution (OSCCR) and penalized by the course staff, at our discretion, up to and including failure of the course.*

Grading Policy

- Homework Assignments (70%)
- Midterm Exam (10%)
- Final Exam (20%)

Final grades will be decided by a "curve fit" policy (with an average around a C/C+ depending on the distribution) that will be announced later in the summer session, after the final homework assignment has been graded. If you have specific concerns about your grades, please contact the instructor and the academic coordinator, as well as your advisor.

Course Summary:

Date	Details	Due
Tue Jul 5, 2022	Synchronous Lecture 1 (https://northeastern.instructure.com/calendar?event_id=858543&include_contexts=course_110667)	9:30am to 11:30am
Wed Jul 6, 2022	Synchronous Lecture 2 (https://northeastern.instructure.com/calendar?event_id=858544&include_contexts=course_110667)	9:30am to 11:30am